

AZAM KHALID

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EDUCATION

Nanyang Technological University, Singapore

Aug. 2024 – May 2028

Bachelor of Computing (Hons) in Computer Science

Singapore

- Dean's List for Academic Year 2025-2026
- Expected Honours (Highest Distinction), Current CGPA: 4.67/5.00

University of Amsterdam

Aug. 2026 – Dec. 2026

Exchange Programme, Computer Science

Netherlands

- Cultural Exchange Ambassador – Led panel discussions on study abroad experiences

EXPERIENCE

Incoming Full Stack Developer Intern

Jan. 2027 – May 2027

Seagate Technologies

Singapore

Undergraduate Research Assistant

Aug. 2025 – July 2026

Nanyang Technological University

Singapore

- Extended SIERRA, a visual Neo4j query interface, by implementing 10 new Cypher features, raising visual read-query coverage from 23% to 69% of standard Cypher clauses using React, Node.js, and Docker
- Built a real-time query translation engine converting visual graph queries into executable Cypher, validated against real-world graph datasets containing over 4,000 graph elements
- Implemented a responsive join diagnostic system that flagged type mismatches and invalid predicate conditions before execution, preventing unnecessary query round-trips to the database
- Authored a research paper proposing novel visual abstractions for graph query interfaces and collaborated with faculty to translate concepts into production-ready software features

PROJECTS

Compendium – Modular AI Context Compiler | Python, MCP, EasyOCR, OpenCV, Docker

June 2026

- Developed a modular document conversion engine normalising 13+ file formats into structured Markdown with chunk-based storage, reducing AI context preparation from manual steps to a single CLI command
- Architected a background daemon that watches directories for changes and continuously compiles supported files into a unified Markdown document, ensuring context remains synchronised with source files
- Built a pluggable OCR pipeline with a four-stage processing workflow and engine-agnostic architecture, enabling seamless integration of multiple OCR engines while maintaining high recognition accuracy
- Engineered an MCP server exposing structured knowledge through reusable resources and tools, enabling LLMs to retrieve semantically organised context instead of processing raw documents

TransitSG | React.js, Vite, JavaScript, Python, FastAPI, PostgreSQL, Unicorn

Sept. 2025

- Devised a transit management platform integrating Singapore's DataMall APIs, reducing commuter decision time by consolidating live crowd density, outage alerts, and route planning into a unified interface
- Implemented a least-crowded route finder using live and forecasted data to minimise exposure to congested stations
- Designed interactive visualisations transforming real-time transit telemetry into congestion insights for commuters
- Led a team of 5 developers using Agile methodologies, Git-based workflows, code reviews, and pull requests

CO-CURRICULAR ACTIVITIES

CCDS Transition Orientation Programme

Nov. 2025 – Aug. 2026

Technology Backend Officer

- Led backend development for large-scale freshman orientation systems by building real-time game infrastructure using Node.js, Socket.IO, PostgreSQL, and AWS (EC2), enabling live synchronisation across multiple games and participants
- Assembled and implemented scalable RESTful APIs in Node.js in collaboration with frontend and design teams, delivering real-time game functionality from product requirements
- Conducted integration, stress, and end-to-end testing to identify bottlenecks and ensure reliable live operation

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Bash, R

Frontend: React, Next.js, HTML, CSS

Backend: Node.js, FastAPI, Flask, REST APIs

Databases: PostgreSQL, MongoDB, Neo4j, Redis

Developer Tools: Git, Docker, GitHub Actions, Linux, AWS, Webpack, Supabase, MCP, Tableau, Vim, RStudio

ACHIEVEMENTS

- picoCTF, Carnegie Mellon University – Ranked 179th out of 8,802 participants worldwide (Top 2%) Mar. 2026
- President Research Scholar, Nanyang Technological University Aug. 2026